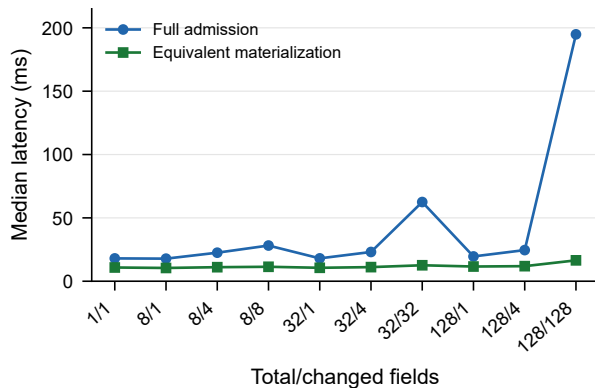
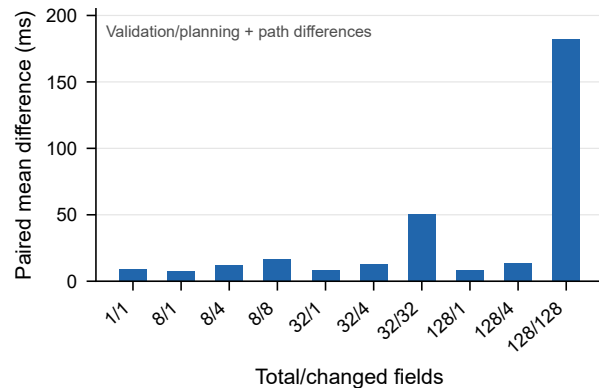
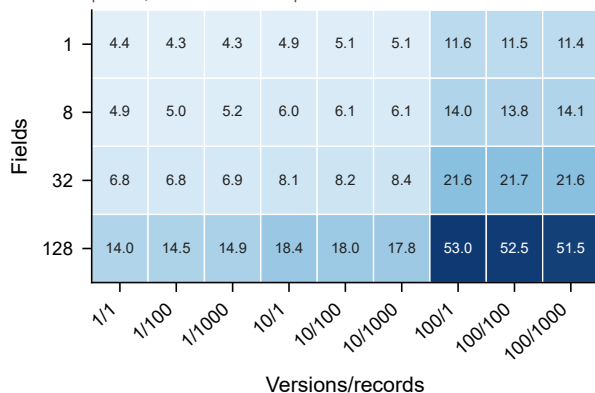
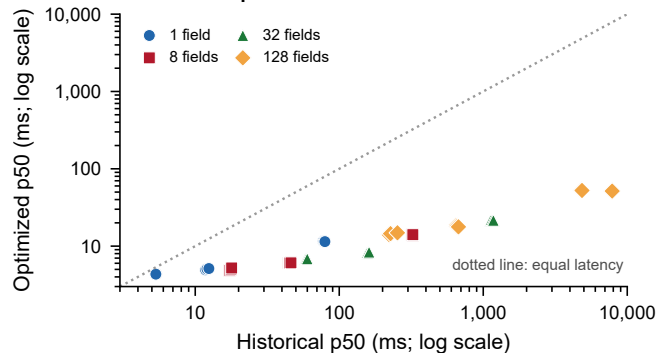


a Equivalent admission comparison**b Paired full-minus-materialization gap****c Optimized reverse-trace latency**

p50 ms; 12 SQL statements per observation

**d Trace comparison: all 36 cells**

Frozen grid: 2,000 admission pairs; 7,200 observations per trace execution.
Trace arms were recorded in separate executions; every frozen cell is shown.